🔹 1. Project Title

Brute Force Attack Detection using Splunk and BOTSv3 Dataset

🔹 2. Introduction

Brute force attacks involve repeated login attempts using multiple password combinations to gain unauthorized access. In this project, log data from the BOTSv3 Dataset is analyzed using Splunk Enterprise to detect suspicious authentication activity.

🔹 3. Objective

- Detect failed login attempts
- Identify suspicious IP addresses
- Analyze brute force attack patterns
- Build detection queries and dashboards

🔹 4. Tools Used

- Splunk Enterprise
- Search Processing Language (SPL)
- BOTSv3 Dataset

🔹 5. Dataset Description

The BOTSv3 dataset contains simulated enterprise logs including authentication logs, network logs, and system activity. It is widely used for cybersecurity training and detection practice.

🔹 6. Implementation Steps

Step 1: Data Ingestion

Uploaded BOTSv3 dataset into Splunk

Verified data using(Query): index=botsv3

Step 2 : Authentication Failure Analysis
Query:

```
 index=* failed OR failure
| stats count by host source
| sort -count
```

This query identifies systems generating the highest number of authentication failures to detect suspicious activity.

Step 3 : Top Attacking Systems

Query:

```
index=* failed OR failure
| stats count by host
| sort -count
| head 10
```

This query highlights the top systems responsible for the highest number of failed login attempts.

Step 4: Targeted Users
Query:

```
index=* failed OR failure
| stats count by username
| sort -count
| head 10
```

This query identifies users who are most frequently targeted by failed login attempts.

Step 5: Brute Force Detection
Query:

```
index=* failed OR failure
| stats count by host source
| where count > 5
| sort -count
```

This query detects potential brute force attacks by filtering systems with multiple failed login attempts above a defined threshold

Step 6 : Brute Force Attack Time Analysis

Query:

```
index=* failed OR failure
| stats count min(_time) as first max(_time) as last by host source
| where count > 10
| convert ctime(first) ctime(last)
```

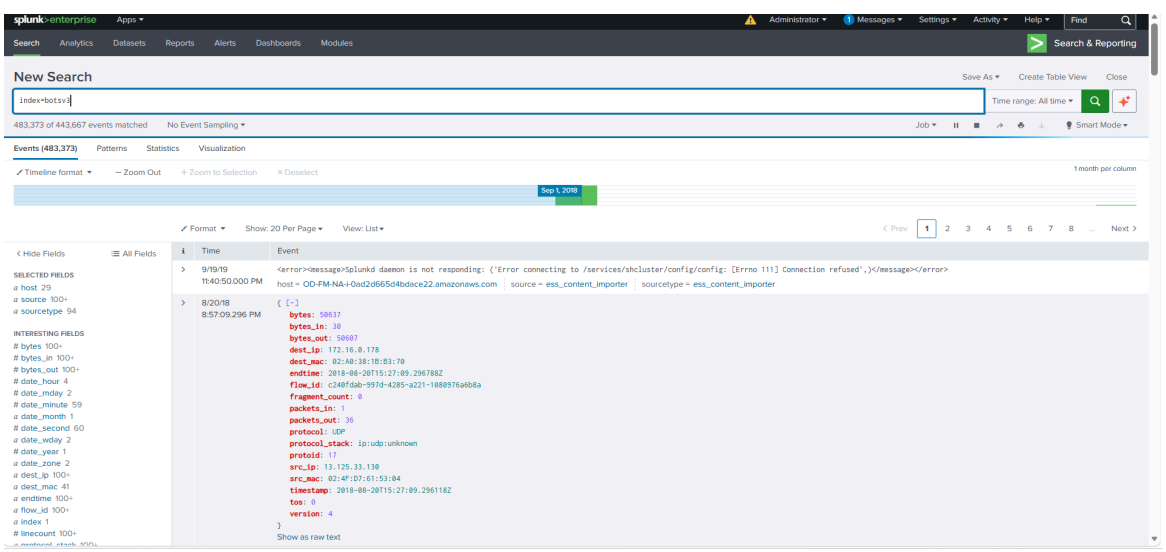
This query analyzes the time duration of brute force attacks by calculating the first and last occurrence of repeated failed login attempts.

Step 7: Dashboard Creation

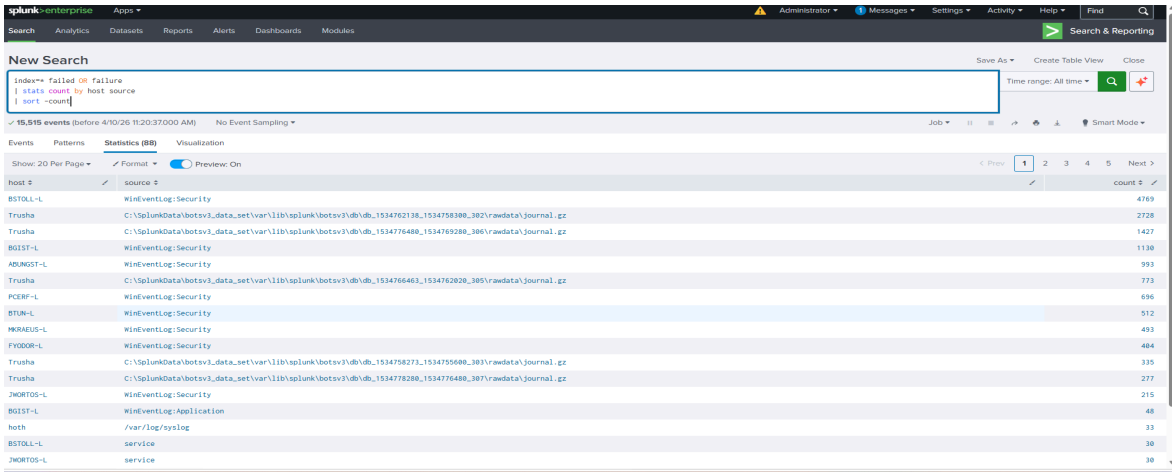
- Created multiple panels to visualize attack patterns
- Used bar charts for comparison of attack sources
- Used tables for detailed analysis
- Used line charts for time-based attack trends

🔹 7. Results

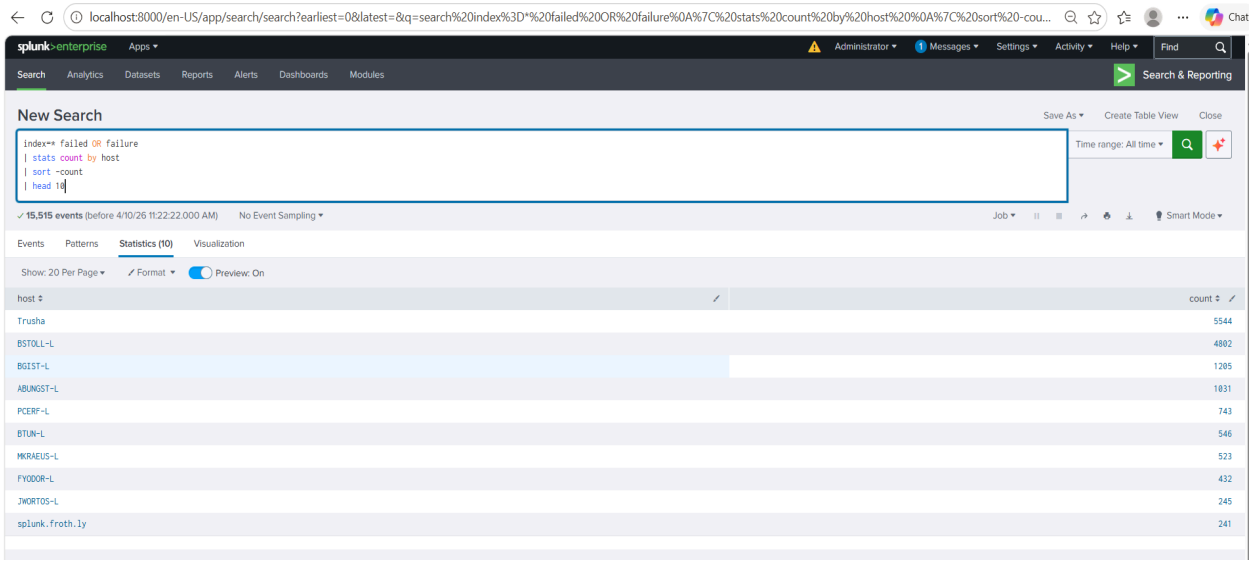
1)BOTSv3 Dataset Loading



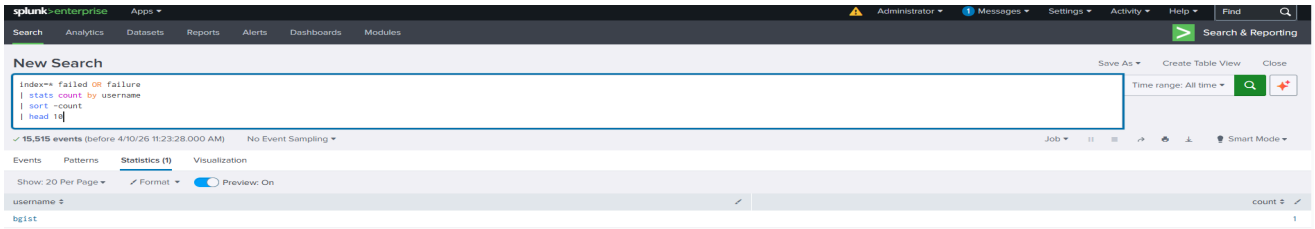
2)Authentication and failure analysis



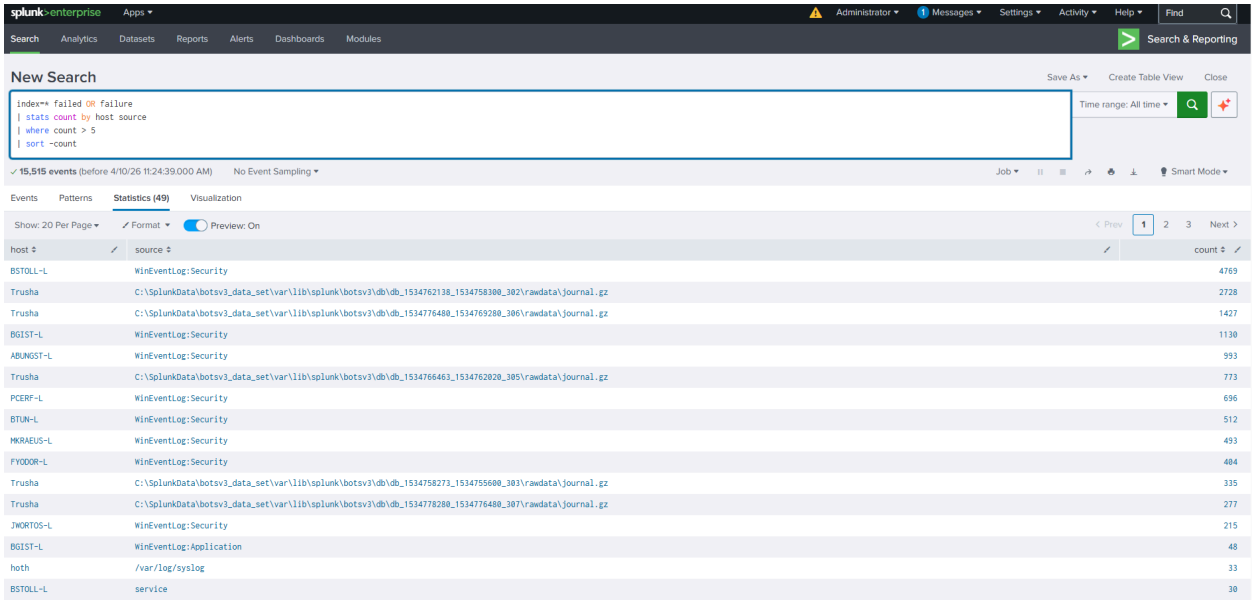
3)Top attacking systems



4) Targeted Users

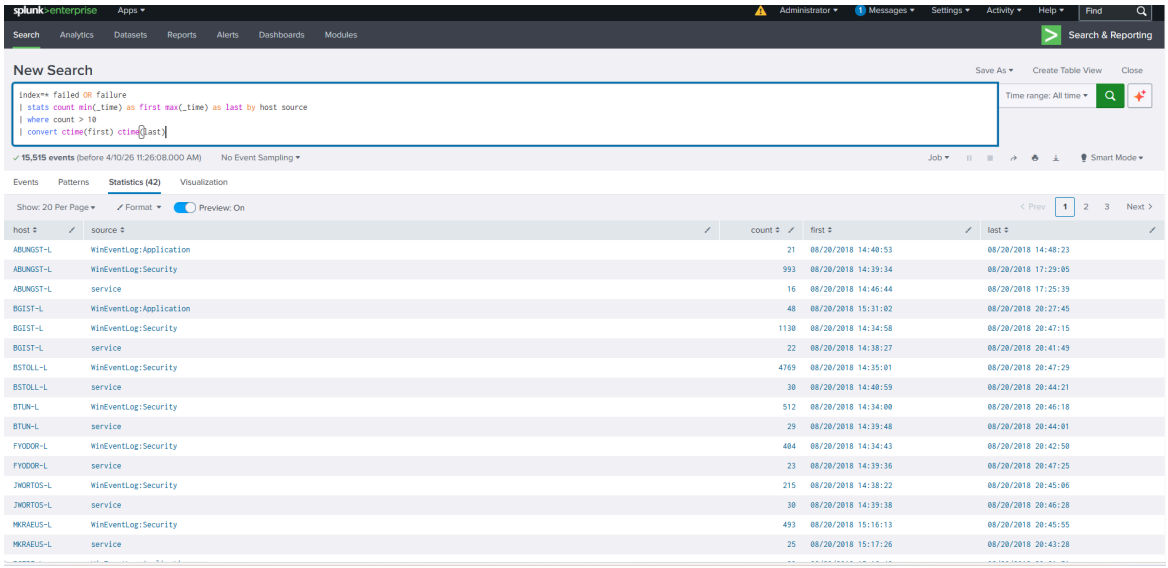


5) Brute Force Detection



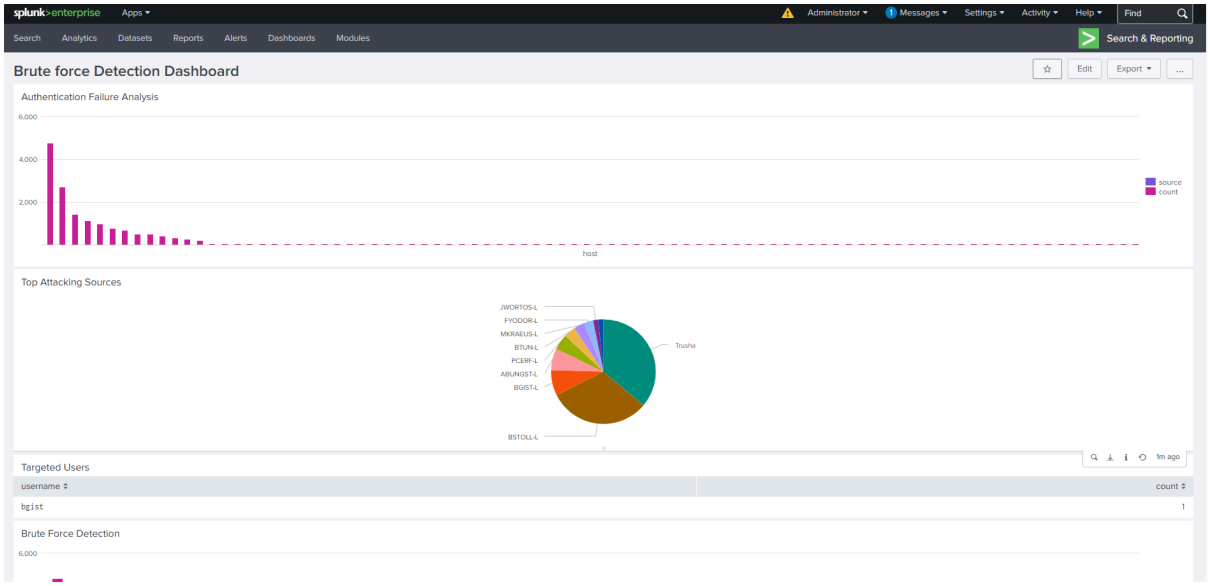
host	source	count
BSTOLL-L	WinEventLog:Security	4769
Trusha	C:\SplunkData\botsv3_data_set\var\lib\splunk\botsv3\db\db_1534762138_1534758300_302\rawdata\journal.gz	2728
Trusha	C:\SplunkData\botsv3_data_set\var\lib\splunk\botsv3\db\db_1534776480_1534769280_306\rawdata\journal.gz	1427
BGIST-L	WinEventLog:Security	1130
ABUNGST-L	WinEventLog:Security	993
Trusha	C:\SplunkData\botsv3_data_set\var\lib\splunk\botsv3\db\db_1534766463_1534762020_305\rawdata\journal.gz	773
PCERF-L	WinEventLog:Security	696
BTUN-L	WinEventLog:Security	512
MKRAEUS-L	WinEventLog:Security	493
FYODOR-L	WinEventLog:Security	404
Trusha	C:\SplunkData\botsv3_data_set\var\lib\splunk\botsv3\db\db_1534758273_1534755600_303\rawdata\journal.gz	335
Trusha	C:\SplunkData\botsv3_data_set\var\lib\splunk\botsv3\db\db_1534778280_1534776480_307\rawdata\journal.gz	277
JWORTOS-L	WinEventLog:Security	215
BGIST-L	WinEventLog:Application	48
hoth	/var/log/syslog	33
BSTOLL-L	service	30

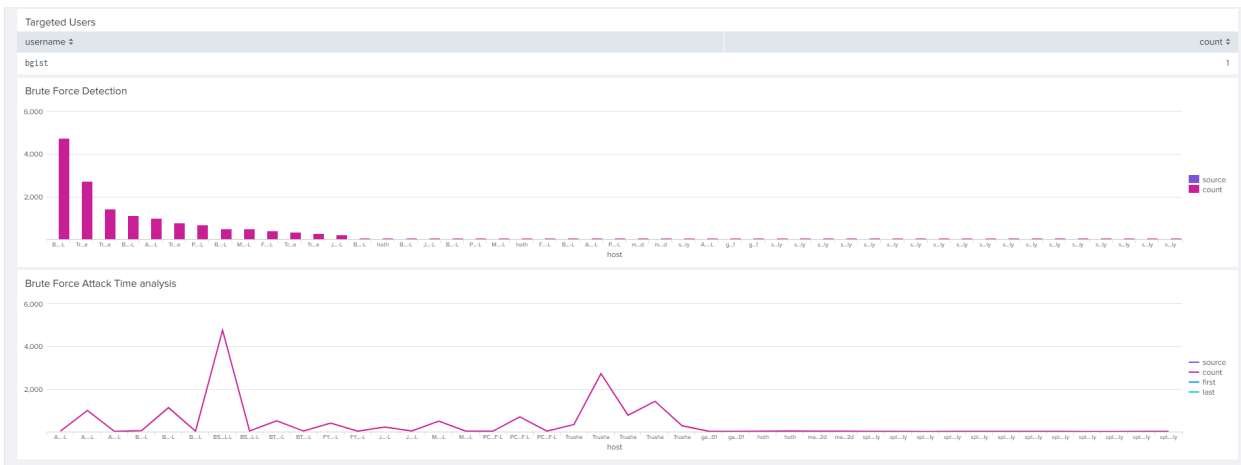
6) Brute Force Attack Time Analysis



host	source	count	first	last
ABUNGST-L	WinEventLog:Application	21	08/20/2018 14:40:53	08/20/2018 14:48:23
ABUNGST-L	WinEventLog:Security	993	08/20/2018 14:39:34	08/20/2018 17:29:05
ABUNGST-L	service	16	08/20/2018 14:46:44	08/20/2018 17:25:39
BGIST-L	WinEventLog:Application	48	08/20/2018 15:31:02	08/20/2018 20:27:45
BGIST-L	WinEventLog:Security	1130	08/20/2018 14:34:58	08/20/2018 20:47:15
BGIST-L	service	22	08/20/2018 14:38:27	08/20/2018 20:41:49
BSTOLL-L	WinEventLog:Security	4769	08/20/2018 14:35:01	08/20/2018 20:47:29
BSTOLL-L	service	30	08/20/2018 14:40:59	08/20/2018 20:44:21
BTUN-L	WinEventLog:Security	512	08/20/2018 14:34:00	08/20/2018 20:46:18
BTUN-L	service	29	08/20/2018 14:39:48	08/20/2018 20:44:01
FYODOR-L	WinEventLog:Security	404	08/20/2018 14:34:43	08/20/2018 20:42:50
FYODOR-L	service	23	08/20/2018 14:39:36	08/20/2018 20:47:25
JWORTOS-L	WinEventLog:Security	215	08/20/2018 14:38:22	08/20/2018 20:45:06
JWORTOS-L	service	30	08/20/2018 14:39:38	08/20/2018 20:46:28
MKRAEUS-L	WinEventLog:Security	493	08/20/2018 15:16:13	08/20/2018 20:45:55
MKRAEUS-L	service	25	08/20/2018 15:17:26	08/20/2018 20:43:28

❖ DASHBOARD





🔹 8. Use Case

This project helps Security Operations Center (SOC) analysts detect unauthorized login attempts and prevent account compromise.

🔹 9. Challenges Faced :

- Dataset ingestion issues
- Writing SPL queries

🔹 10. Future Improvements

- Add alerts in Splunk
- Improve dashboards

🔹 11. Conclusion

This project demonstrates how brute force attacks can be detected using log analysis in Splunk by identifying repeated failed login attempts and suspicious patterns.